

The FFN Flip is Peak-Relative: A Cross-Family Anatomy of Truth Erosion in Small Language Models

A working note: three models, two families, one sequence — build, peak, flip

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Abstract

This working note closes the anatomy chapter opened in “Anatomy of the Truth Axis in Qwen2.5-1.5B” and extends it across scale and family. The central finding, confirmed by pre-registered prediction on three models spanning two families (Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-1B), is that the feed-forward network’s role reverses at a fixed *peak-relative* position: the FFN writes *pro*-truth into the layer where the one-dimensional truth axis peaks, and flips to writing *anti*-truth at exactly peak+1 — regardless of absolute depth, relative depth, model size, or family. Aligned in peak-relative coordinates, the three per-layer profiles of the FFN’s truth-axis writing collapse onto a common shape (Fig. 1). Inside the SwiGLU, an exact algebraic split attributes the anti-truth writing to the *value stream* (up-projection), not the gate: the specific “context-switching” hypothesis is falsified, causally and observationally. A competing static hypothesis — that the gauge-invariant OV/QK circuit crossover of the attention weights predicts where truth lives — is falsified by a pre-registered criterion: across the two Qwen scales the crossover and the peak move in *opposite* directions in relative depth. The knowledge-dependent dimensionality law of the companion paper gains its first cross-family points: the axis-vs-probe gap tracks a behaviorally measured knowledge rate (gap 0.049 at high knowledge, 0.141 at mid, ≈ 0.20 at low). Finally, in the mid-knowledge regime a second readable dimension emerges (+0.092 held-out AUC over the single axis), refining — not contradicting — the one-dimensionality claim, which was always scoped to known facts. All results carry the shields of the companion papers: held-out cross-validation over pairs, permutation nulls, exact identity checks, and pre-registered predictions; retractions are reported as results.

1 Scope and honesty statement

Nothing here has the status of a settled result; everything is labeled. Results are organized in three evidential levels and never mixed: **Level A** — replicated across models with predictions written before the runs; **Level B** — single-model but causal, with multiple independent methods agreeing; **Level C** — leads: single seed, open confounds, or margins at threshold, declared as such. Three items from earlier stages of this work are *retracted or falsified* below and reported in full: (i) the “attention builds, FFN consolidates” story (already overturned in the previous note); (ii) the hypothesis that the SwiGLU *gate* carries the erosion (falsified here in favor of the value stream); (iii) the hypothesis that the static OV/QK crossover predicts the truth peak (falsified here by pre-registered criterion). A fourth item, the “FFN-off beats frozen” margin, failed to replicate across scale and is demoted from lead to noise. The reader following the repository can reproduce every number with the commands of Section 8.

2 Setup

Three base (non-instruct) models:

model	blocks	d_{model}	Q/KV heads	truth peak p	rel. depth
Qwen2.5-1.5B	28	1536	12/2	15	0.54
Qwen2.5-3B	36	2048	16/2	16	0.44
Llama-3.2-1B	16	2048	32/8	7	0.44

Dataset: NeelNanda/counterfact-tracing (revision c945b082..., 21,919 rows), 250 minimal pairs, loaded as Parquet only. The unsupervised truth axis v_1 , the pairing design, and the held-out protocol are exactly those of the companion paper. Conventions: “block b ” denotes the output of transformer block b , i.e. `hidden_states[b+1]`; the “peak” p is the block whose intact residual maximizes held-out truth AUC. All decomposition runs use float32 and embed the additive identity check of the previous note (median relative error 1.06×10^{-7} / 1.17×10^{-7} / 6.52×10^{-8} on the three models); a run is certified by its in-code checks, not by its launch conditions. Extraction-only runs (peak finding, behavioral checks) use bfloat16, as in the companion paper.

3 Methods added in this note

Fixed-axis contribution gap. The previous note fit the axis per component and per layer; sign and rotation ambiguities of refitting motivated a stricter instrument. Here v_1 is fit *once*, on the intact residual at the peak block, on training folds only, oriented there (true = positive), and every attention and FFN contribution at every layer is projected onto that same fixed direction. The statistic is the held-out class gap of a contribution c , $\text{gap} = \overline{v_1 \cdot c} | \text{true} - \overline{v_1 \cdot c} | \text{false}$, and its effect size $d' = \text{gap}/\text{pooled std}$. A negative FFN gap means the FFN literally writes toward the false side of the very axis on which the model represents truth.

Exact SwiGLU split. The FFN computes $f = W_{\text{down}}(g \odot u)$ with $g = \text{silu}(W_{\text{gate}}x)$, $u = W_{\text{up}}x$, elementwise in the shared expanded basis. The projection of f on v_1 is $w \cdot (g \odot u)$ with $w = W_{\text{down}}^\top v_1$: the residual axis pulled back through the down-projection. *No axis is ever fit in the 8960/11008-dimensional space*, avoiding the $p \gg n$ regime entirely. With the symmetric intra-pair split ($\Delta g = g_t - g_f$, $\bar{g} = (g_t + g_f)/2$) the mixed term cancels and the identity is exact:

$$w \cdot (g_t \odot u_t - g_f \odot u_f) = \underbrace{w \cdot (\Delta g \odot \bar{u})}_{\text{gate term}} + \underbrace{w \cdot (\bar{g} \odot \Delta u)}_{\text{value term}}. \quad (1)$$

Verified numerically at 7.4×10^{-8} relative error on real data.

Interventions. Beyond zero-ablation over a band, two refinements: (i) a *frozen control* obtained for free — with both attention and FFN zeroed over the band, the readout state is *identically* the band-entry state, so the frozen reference is read off the intact pass; (ii) *gate-freeze* / *value-freeze*: at the reading position, g (resp. u) is replaced by its intra-pair mean cached from the intact run, killing the class-driven variation of one stream while the FFN stays on. Zeroing the gate would zero the whole FFN and test nothing new.

Static circuit norms. Per head (GQA-aware: each group of query heads shares one K/V), the gauge-invariant circuit norms $\|W_q^\top W_k\|_F$ (routing) and $\|W_o W_v\|_F$ (writing) [4]. Individual weight norms are gauge-dependent ($W_q \rightarrow \alpha W_q$, $W_k \rightarrow W_k/\alpha$ leaves the function unchanged); the products are not. A gauge self-test (rescale and verify invariance at machine precision) gates the computation.

Behavioral label, quantitative. Section 9 of the companion paper (“verify behavior before geometry”) is made a number: greedy-decode $n = 40$ CounterFact prompts and measure the fraction where the true target is produced (a rough lower bound). The KNOWN rate is recorded *before* the geometry is measured and pre-registers the expected axis-vs-probe gap.

4 Level A: replicated, pre-registered

4.1 The FFN flip is peak-relative

For each model, the prediction was written before the run: *the FFN’s fixed-axis gap is pro-truth at the peak block p and flips to stable anti-truth at $p + 1$* . Table 1 shows the measured values around the flip; “stable” uses the criterion of the previous consolidation ($|\text{mean}| > \max(2\sigma_{\text{seeds}}, 0.03)$, 5 CV seeds).

Table 1: The flip at peak+1, fixed axis at the peak, CounterFact 250 pairs. Gap is mean $\pm$ std over 5 seeds where consolidated (Qwen2.5-3B, Llama-3.2-1B); Qwen2.5-1.5B values are from the single-seed scan of the previous session, consistent with its 5-seed consolidation. Permutation null: 100 within-pair swaps, two-sided.

model	layer	gap _{ffn}	d'_{ffn}	perm. p	verdict
Qwen2.5-1.5B	15 ($= p$)	+0.728	+0.57	—	pro-truth
	16 ($= p+1$)	−0.692	−0.82	—	anti-truth (flip)
Qwen2.5-3B	16 ($= p$)	$+0.575 \pm 0.005$	+0.68	—	pro-truth stable
	17 ($= p+1$)	-0.926 ± 0.002	−1.25	0.0099	anti-truth stable (flip)
Llama-3.2-1B	7 ($= p$)	$+0.092 \pm 0.001$	+0.54	—	pro-truth stable
	8 ($= p+1$)	-0.111 ± 0.001	−0.94	0.0099	anti-truth stable (flip)

The three peaks sit at different absolute depths (15, 16, 7) and different relative depths (0.54, 0.44, 0.44); the flip sits at peak+1 in all three. The architecture does not fix *where*; it fixes the *sequence*: build $\rightarrow$ peak $\rightarrow$ flip at the next block. Figure 1 aligns the full profiles in peak-relative coordinates; normalized by the flip amplitude, the three curves share one shape — rise into the peak, spike at +1, decay toward zero. The collapse is not perfect and the imperfections are data: the 1.5B pre-peak side oscillates where the other two rise monotonically, and the post-flip decay profiles differ (the 3B erosion band is non-monotone: −1.25, −0.14, −0.76 at layers 17–19).

4.2 The dimensionality law gains cross-family points

The companion paper’s most novel claim — the truth representation is near one-dimensional when the model knows the fact and diffuse when it does not, quantified by the gap between the unsupervised axis and a full linear probe — was measured on one model. It now has three points on two families, with the knowledge level measured behaviorally where pre-registration required it:

model	KNOWN rate	axis AUC	probe AUC	gap
Qwen2.5-3B	(high; not measured, see text)	0.832	0.881	0.049
Llama-3.2-1B	42% (17/40, measured first)	0.690	0.831	0.141
Qwen2.5-1.5B	(low–mid; companion paper)	0.638–0.795	0.841	≈ 0.20 (mix)

For Llama-3.2-1B the expectation (“mid knowledge $\rightarrow$ intermediate gap”) was registered from the 42% KNOWN rate *before* the probe was run; the measured gap of 0.141 fell inside the registered window between the known-fact regime (≈ 0.05) and the unknown-fact regime (≈ 0.20). Making the table fully uniform requires running the same behavioral check on the two

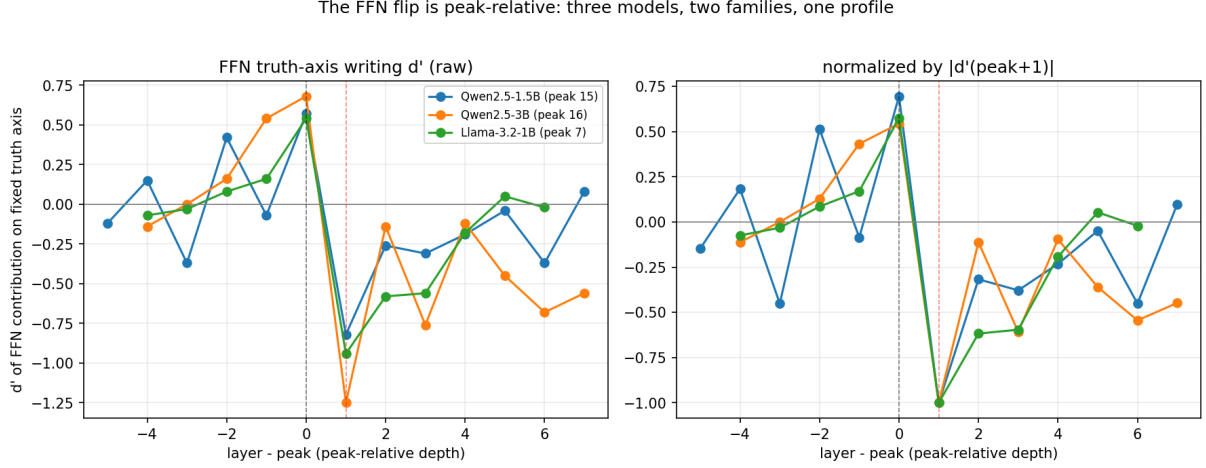


Figure 1: The FFN’s truth-axis writing (d' of the fixed-axis contribution gap) across depth, aligned at the truth peak (dashed black) — three models, two families. Right: normalized by the flip amplitude $|d'(p+1)|$. The red dashed line marks peak+1.

Qwen models; this is rerun-cheap and left as the first item of the next chapter’s prerequisite (the KNOWN/UNKNOWN behavioral label).

4.3 Pre-registered negative: the static OV/QK crossover does not track the peak

Motivated by an external observation that value-to-query weight-norm ratios scale with depth across architectures [6], we tested the gauge-invariant version: does the layer where the writing circuit $\|W_o W_v\|$ overtakes the routing circuit $\|W_q^\top W_k\|$ (median-normalized, sustained for 3 consecutive layers — criterion fixed in code before the runs) predict the truth peak? On Qwen2.5-1.5B the crossover (layer 17) sits one block from the flip — consistent. The cross-scale prediction registered before the second run discriminated two hypotheses: peak tracks crossover (3B peak at $\sim 22 \pm 2$) vs. fixed relative depth (peak at ~ 19). The measured 3B peak is block 16, eight blocks from its crossover (24); across the two scales the crossover moves *deeper* in relative depth ($0.61 \rightarrow 0.67$) while the peak moves *shallower* ($0.54 \rightarrow 0.44$). Two points moving in opposite directions falsify the link. The depth-scaling of the circuits themselves is visible in our tables and may well be real and general; it is simply not where truth lives. The late-layer OV explosion (e.g. $\|W_o W_v\|$ sums of 338.9/322.0 at Qwen-1.5B layers 26–27) instead matches the *decoupling* of Section 6.

5 Level B: causal, single-model, three methods agreeing

5.1 The value stream, not the gate, writes the erosion (Qwen2.5-1.5B)

The working hypothesis entering this chapter was contextual: the SwiGLU *gate* (which neurons open) switches the stream toward next-token/format context and thereby erodes truth. Three independent methods return the same verdict, and it is the opposite attribution:

layer	gate term	value term	total	gate share
13	+0.249	+0.099	+0.348	0.72
15 (p)	+0.486	+0.242	+0.728	0.67
16 ($p+1$)	+0.041	− 0.734	−0.692	−0.06
17	+0.123	−0.385	−0.262	−0.47
18	−0.004	−0.279	−0.284	0.02

(i) The exact split of Eq. 1: at the flip the anti-truth total is carried almost entirely by the value term; the gate term stays neutral or mildly pro-truth (at layer 17 the two streams *fight*, and the gate loses). Notably the same split explains the pro-truth construction at the peak: two thirds gate. (ii) Causal freezes over the band (readout block 18): value-freeze recovers +0.089 AUC (0.801, essentially the frozen reference 0.793); gate-freeze recovers only +0.040. (iii) A qualitative logit-lens of the band FFN contributions shows format/template content (punctuation, scaffolding fragments, code tokens) dominating both classes, with both classes pushed toward the false side of the axis and true statements pushed harder (layer 16: mean projection -2.36 true vs. -1.65 false). The specific gate/context hypothesis is falsified; the corrected statement is: *after the peak, the FFN’s value stream injects prediction-oriented content whose along-axis component is systematically anti-truth.*

5.2 The rotation objection, raised and closed

An objection raised mid-session (by the author, against the author): could the negative gap be a rotating write-direction seen through a fixed axis — a projection artifact — rather than genuine anti-truth writing? Rotation makes a precise prediction: the cosine of the mean FFN pair-difference with v_1 should sweep smoothly through zero across several layers with roughly constant magnitude. The data show the opposite on both consolidated models: a one-step sign jump at peak+1 with the magnitude intact (Qwen-3B: $\cos = +0.245 \rightarrow -0.409$, $\Delta = -0.654$, $\|d\|$ 2.52→2.34; Llama-1B: $+0.292 \rightarrow -0.341$, $\Delta = -0.632$, $\|d\|$ 0.33→0.34). It is erosion, not rotation. A secondary reading of the same table: most of what the FFN writes is *orthogonal* to the axis (class-blind volume), with a small, real, systematically anti-truth along-axis component on top.

5.3 Corrected ablation, the frozen control, and a demoted lead

The previous note’s ablation refit the axis on ablated states; this note adds the fixed-axis readout and the frozen control. On both models, zeroing the FFN over the erosion band raises truth AUC at the readout while zeroing attention does not raise it (Qwen-1.5B band 16–18: intact 0.712, FFN-off 0.815, attn-off 0.708, frozen 0.793; Qwen-3B band 17–19: intact 0.793, FFN-off 0.846, attn-off 0.738, frozen 0.831), and the axis-geometry cosines show it is the FFN that rotates the readable direction (intact entry→readout $|\cos|$ 0.627/0.667 vs. 0.876/0.890 with the FFN off). Two honest annotations. First, the earlier lead “FFN-off beats the frozen reference” did *not* replicate: +0.022 (1.5B) vs. +0.015 (3B), both straddling the noise threshold; the supported statement is FFN-off $\approx$ frozen on both models — no robust evidence of recoverable signal beyond the peak. Second, a scale difference: in the 3B, attention-off in the band *hurts* (-0.055), consistent with its post-peak attention still writing pro-truth (fixed-axis gap +0.663 at block 18): the 3B erosion band is a tug-of-war, not a passive decline.

6 Level C: leads, declared as leads

A second readable dimension in the mid-knowledge regime. On Llama-3.2-1B at the peak, a held-out logistic on (Re, Im) reaches 0.783 vs. 0.690 for the single axis (+0.092); the phase score alone reaches 0.755. In the Qwen models the second dimension never added more than +0.02. The companion paper contained a faint precursor (mix data: PHASE 0.676 vs. 1D 0.638) that was retracted via the sign-echo argument; that argument explains why phase *separates* but does not bound it *below* the axis, and a logistic that contains the 1D model as a special case generalizing +0.092 better is stronger evidence. Read through the dimensionality law, this is the expected geometric signature of the mid-knowledge regime: as knowledge decreases, truth de-concentrates and off-axis components become readable. It refines the one-dimensionality claim,

which was always scoped to known facts. Status: one model, one seed battery, no dedicated null on the 2D–1D margin yet — a lead.

The ignition step at scale. The 1.5B per-layer AUC curve is a smooth ramp (used in the companion paper to falsify “departments”). The 3B curve has a step: $0.526 \rightarrow 0.790$ in a single layer, driven (per the fixed-axis gaps) by a coordinated two-layer construction burst in which both components write pro-truth strongly (attention $d' + 1.58/ + 1.27$ at blocks 15–16). Truth in the 3B does not grow; it switches on. If it survives re-seeding, this is a qualitative difference across scale and reopens the departments question exactly where the companion paper closed it.

Framing bias with an open lexical confound. A 2×2 moral/prudential design (24 quads; e.g. “X is wrong because it harms others” vs. “X is risky because it can lead to prison”) on Qwen2.5-1.5B: unlike polarity, framing does *not* rotate the axis (moral axis transfers to prudential at 0.979–0.99; no flip), but there is a large level bias among equally-true statements (-2.75σ at layer 11, stable $\approx -2.5\sigma$ through the middle band), and the unsupervised SVD of mixed-framing pairs recovers a framing direction at $|\cos| = 0.912$ in PC2 (with t_G in PC1 at 0.993) — the same structure as the polarity recovery of the companion paper, on a new factor. The confound: the bias is already -2.86σ at layer 1, which shouts lexical surface (“wrong” vs. “risky”), not deep context. Crossed-template controls are required before this is more than a lead.

Late-layer decoupling. In all inspected models the total volume written by the FFN grows toward the output (contribution norms 16→38 on the 1.5B; OV circuit norms exploding in the last blocks) while its projection on the truth axis goes to zero and the intact truth AUC decays to near chance. The terminal regime is not an oscillation of the axis but a *decoupling*: the model writes enormously, and none of it speaks of truth. Small terminal sign changes in the gap are zero-crossings of a dying signal (below both the stability criterion and the permutation null) and must not be read as structure.

Future work: per-head anatomy at the peak. Throughout this note the attention contribution is treated as a sum; it is in fact the sum of 12/16/32 heads (per model) through the output projection, and the fixed-axis instrument of Section 3 applies unchanged to each head’s slice $W_o[:, h] \cdot \text{head}_h$. The open question is whether the pro-truth construction at the peak (and the post-peak tug-of-war of the 3B) is carried by a few identifiable heads — which would make it causally ablatable and connect this anatomy to the head-specialization literature [6] — or distributed across the pack. The extraction is mechanical (a GQA-aware pre-hook on the output projection, mirroring the down-projection hook of the SwiGLU analysis) and is left as the opening experiment of the next chapter, to be run with the same seed/null shields as the rest of this note.

7 Limitations

Three models are not many, and both Qwen models share a training recipe; the cross-family evidence rests on one Llama at 1B. The behavioral KNOWN rate is a string-match lower bound on 40 prompts and was measured on one of three models. The flip’s *position* replicates; the erosion *profile* does not (non-monotone in the 3B), and nothing here explains *why* the flip is pinned to peak+1 — the note measures the sequence, not its cause. The framing result carries an explicit lexical confound. The value-stream attribution (Level B) is causal on one model only; the corresponding runs on the 3B and the Llama are mechanical to add. Ablation measures necessity for linear readability of truth, not for behavior. The gate/value freezes act at the reading position only. None of this has undergone a falsification battery of the scale applied to the companion paper’s central claim.

8 Reproducibility

Scripts (reusing `truth_probe.py` and `anatomy.py` from the companion repository; every tool embeds its identity / null / stability shields): `ffn_erosion.py` (subcommands `contrib`, `ablate`, `sweep`, `lens`), `swiglu.py` (`attrib`, `framing`, `gatefreeze`), `circuits.py`, `flip_consolidate.py`, `behav_check.py`. Dataset revision is pinned in Section 2; models load with safetensors only and `trust_remote_code=False`. Llama-3.2 requires accepting the license on Hugging Face and a read token. Representative commands:

```
# behavioral label BEFORE geometry (Sec. 9 discipline)
python behav_check.py --model meta-llama/Llama-3.2-1B

# find the peak (extraction in bf16, as in the companion paper)
python truth_probe.py signal --dataset counterfact --max-pairs 250 \
    --model meta-llama/Llama-3.2-1B --baseline --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# consolidate the flip: 5 seeds + permutation null + rotation check (fp32)
python flip_consolidate.py --model meta-llama/Llama-3.2-1B \
    --axis-block 7 --flip-layer 8 --scan-start 3 --scan-end 13

# corrected causal ablation with fixed-axis and frozen controls
python ffn_erosion.py ablate --model Qwen/Qwen2.5-3B \
    --band-start 17 --band-end 19 --readout 19

# exact gate/value attribution and causal freezes (Qwen2.5-1.5B)
python swiglu.py attrib
python swiglu.py gatefreeze

# gauge-invariant static circuit norms (weights only, CPU is fine)
python circuits.py --model Qwen/Qwen2.5-3B --dtype bfloat16
```

Upload `flip_collapse.png` alongside this `.tex` so Fig. 1 renders. License: CC BY 4.0.

References

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